A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 1: Information Representation

Subtopic 1.2.2: Multimedia: Sound

Concept 1.2.2.2: Impact of Sampling on File Size and Accuracy

Total Questions: 10

Mark Schemes begin on Page 6

Question 2 | Source: 9618_s24_qp_11 (Q2.d)

2 A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:

- The digital camera in the doorbell starts recording a video.
- A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell's internal secondary storage device and overwritten when the secondary storage device is full.

(d) The digital camera has a microphone which is used to record the sound for the video.

The user changes the sampling rate that the microphone uses from 44.1 kHz to 88.2 kHz.

Describe how this change in sampling rate will affect the performance of the video doorbell.

....

....

....

....

....

.... [3]

Question 3 | Source: 9618_w23_qp_12 (Q6.b)

(b) Explain the impact of changing the sampling resolution on the accuracy of a sound recording.

....

....

....

....

....

.... [3]

Question 4 | Source: 9618_s23_qp_13 (Q3.c.ii)

3 A mobile telephone is used to record a video.

(c) The video includes a sound recording.

(ii) A second video is recorded. The sound in the second video needs to be more precise.

Explain the reasons why increasing the sampling rate and the sampling resolution will improve the precision of the second recording.

Sampling rate ....

....

....

....

Sampling resolution ....

....

....

....
[4]

Question 5 | Source: 9618_w22_qp_11 (Q1.d.ii)

(d) Sound samples are recorded and saved in a file.

(ii) Explain the effect of increasing the **sampling resolution** on the sound file.

....

....

....

.... [2]

Question 6 | Source: 9618_w22_qp_12 (Q6.b.i)

(b) The student records a sound file.

(i) Explain the effect of increasing the sampling rate on the accuracy of the sound recording.

....

....

....

.... [2]

Question 7 | Source: 9618_w22_qp_12 (Q6.b.ii)

(b) The student records a sound file.

(ii) Explain the effect of decreasing the sampling resolution on the file size of the sound recording.

....

....

....

.... [2]

Question 8 | Source: 9618_w22_qp_13 (Q1.a)

1 A digital audio message needs to be recorded.

(a) Tick (✓) **one** box in each row to identify the effect of each action on the accuracy of the recording.

Action	Accuracy increases	Accuracy decreases	Accuracy does not change
Change the sampling rate from 40 kHz to 60 kHz.			
Change the duration of the recording from 20 minutes to 40 minutes.			
Change the sampling resolution from 24 bits to 16 bits.			

[2]

Question 9 | Source: 9618_w22_qp_13 (Q1.b)

1 A digital audio message needs to be recorded.

(b) The audio message is recorded with a sampling rate of 50 kHz and a sampling resolution of 16 bits.

The recording is 20 minutes in length.

Calculate the file size of the recording.

Give your answer in megabytes **and** show your working.

Working ....

....

....

....

Answer .... megabytes

[2]

Question 10 | Source: 9618_w21_qp_11 (Q7.a.i)

7 Bobby is recording a sound file for his school project.

(a) He repeats the recording of the sound several times, with a different sample rate each time.

(i) Describe the reasons why the sound is closer to the original when a higher sample rate is used.

....

....

....

.... [2]

Question 11 | Source: 9618_w21_qp_11 (Q7.a.ii)

7 Bobby is recording a sound file for his school project.

(a) He repeats the recording of the sound several times, with a different sample rate each time.

(ii) Describe the reasons why the sound file size increases when a higher sample rate is used.

....

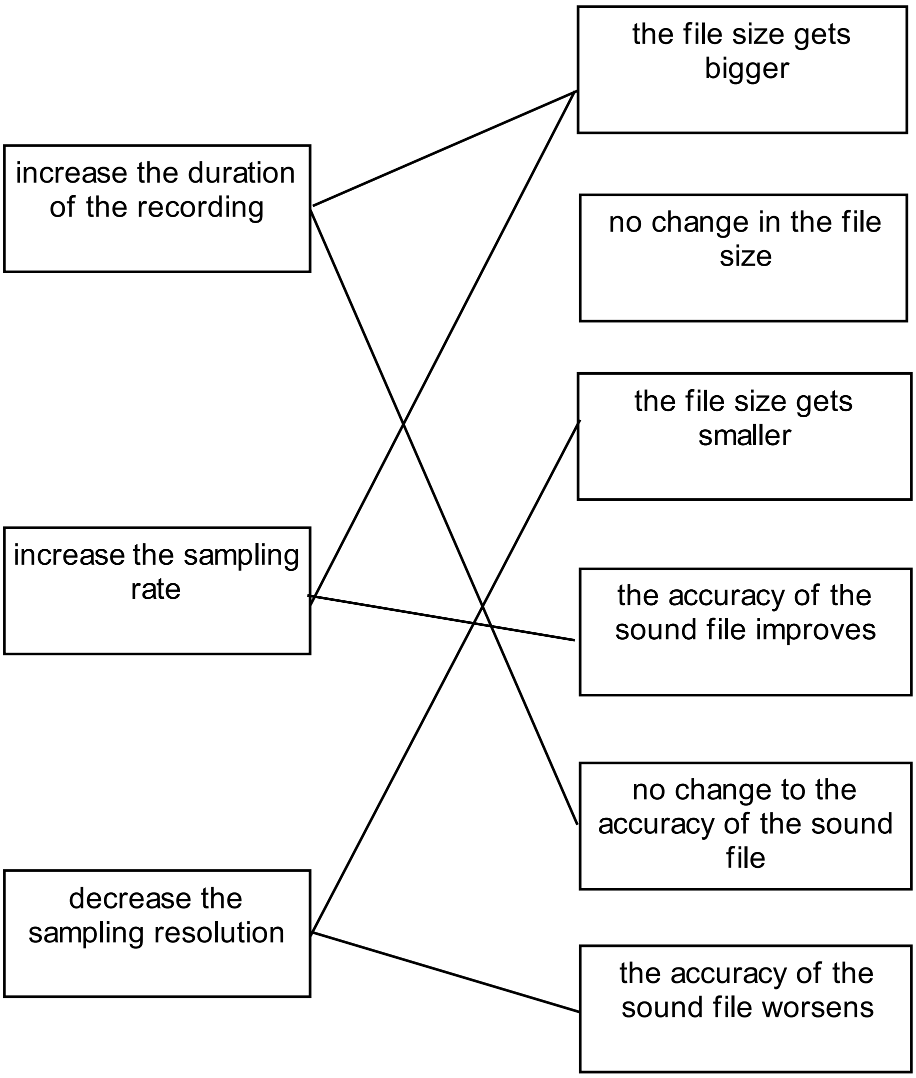
....

....

.... [2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_13 (Q1.a)

<table>
<tr><td>1(a)</td><td>1 mark for each correctly connected change box, max 3 marks

</td><td>3</td></tr>
</table>

Mark Scheme for Question 2 | Source: 9618_s24_ms_11 (Q2.d)

<table>
<tr><td>2(d)</td><td>1 mark each to max 3:

Data transmission to user's smartphone will take longer
… because there is more data to transmit

The secondary storage device will fill faster
… fewer videos will be able to be stored long-term // videos are overwritten more often
</td><td>3</td></tr>
</table>

Mark Scheme for Question 3 | Source: 9618_w23_ms_12 (Q6.b)

6(b)	**1 mark** for each bullet point (**max 3**) Increase sampling resolution • … the number of bits used for each sample is increased • … there will be more values available to represent each sample // more amplitudes can be represented • ... each binary amplitude/note in the digital recording is closer to the analogue amplitude/note • … quantisation errors are reduced • … the digital soundwave is closer to the original analogue soundwave Decrease sampling resolution • … the number of bits used for each sample is decreased • … there will be fewer values available to represent each sample // fewer amplitudes can be stored • ... each binary amplitude/note in the digital recording is further from the analogue amplitude/note • … quantisation errors are increased • … the digital soundwave is less like the original analogue soundwave	**3**

Mark Scheme for Question 4 | Source: 9618_s23_ms_13 (Q3.c.ii)

3(c)(ii)	**1 mark** each; **max 2** for rate and **max 2** for resolution Sampling rate • There are smaller 'gaps' in the sound wave // sound is recorded **more** often • Digital **waveform** is closer to the analogue **waveform** • The quantisation errors are smaller Sampling resolution • There are **more** bits per sample // a wider range of amplitudes can be stored • Each binary **amplitude**/note (in the digital recording) is closer to the analogue **amplitude**/note • Digital **waveform** is closer to the analogue **waveform** • The quantisation errors are smaller	**4**

Mark Scheme for Question 5 | Source: 9618_w22_ms_11 (Q1.d.ii)

1(d)(ii)	**1 mark** for each bullet point (**max 2**): • increases the number of bits per sample // larger range of values • which means that the file size increases • makes the sound file more accurate //digital waveform closer to original (analogue) waveform • smaller quantisation errors	**2**

Mark Scheme for Question 6 | Source: 9618_w22_ms_12 (Q6.b.i)

6(b)(i)	**1 mark** for each bullet point **(max 2)**: • **improves** the accuracy of the sound file • ... because (digital) waveform more closely resembles the **analogue wave**form • quantization errors are reduced • increases the amount of detail stored	**2**

Mark Scheme for Question 7 | Source: 9618_w22_ms_12 (Q6.b.ii)

6(b)(ii)	**1 mark** for each bullet point: • decreases the file size of the sound file • ... because fewer bits are used to store **each sample**	**2**

Mark Scheme for Question 8 | Source: 9618_w22_ms_13 (Q1.a)

1(a) — **1 mark** for **one or two** correct ticks, **2 marks** for **three** correct ticks. — **2**

Action	Accuracy increases	Accuracy decreases	Accuracy does not change
Change the sampling rate from 40 kHz to 60 kHz.	✓		
Change the duration of the recording from 20 minutes to 40 minutes.			✓
Change the sampling resolution from 24 bits to 16 bits.		✓	

Mark Scheme for Question 9 | Source: 9618_w22_ms_13 (Q1.b)

1(b) — **1 mark** for answer; **1 mark** for working. — **2**

Working:

Size = 50KHz * (20 × 60) * 16 bits
= 50 000 * 1200 * 16 bits // 50 000 *1200 * 2 bytes
= 960 000 000 bits
= 120 000 000 bytes
= 120 000 kilobytes
= 120 megabytes

Answer = 120 megabytes

Mark Scheme for Question 10 | Source: 9618_w21_ms_11 (Q7.a.i)

7(a)(i)	**1 mark** per bullet point • Smaller time gaps between the samples • Makes the **digital** sound **wave** more accurate • Smaller quantisation errors	**2**

Mark Scheme for Question 11 | Source: 9618_w21_ms_11 (Q7.a.ii)

7(a)(ii)	**1 mark** per bullet point • More samples/data are taken/recorded • … so more bits are stored altogether	**2**